

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

A. It activates TLR5 on the surface of macrophages.

B. It primarily induces the Th1-cell response.

C. Peptidoglycan is its major antigen that induces an immune response.

D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<reasoning process here> The patient presents with symptoms consistent with community-acquired pneumonia (CAP), including sore throat, fever, non-productive cough, oropharyngeal erythema, and rales on auscultation. The chest X-ray shows patchy reticular opacities in the perihilar regions, which are typical findings in CAP. Given the patient's occupational history as a poultry worker, this raises suspicion for a pathogen commonly associated with that environment—particularly avian origin infections such as *Mycoplasma pneumoniae*.

Mycoplasma pneumoniae is the most common cause of CAP and is known for its ability to induce a weak inflammatory response due to the lack of a cell wall. It does not contain peptidoglycan (which would be present in bacteria with cell walls), making option C incorrect because peptidoglycan is not its major antigen. Instead, *M. pneumoniae* has a cell membrane made of lipoproteins and other components that trigger immune responses via TLRs (Toll-like receptors), including TLR2, not TLR5. TLR5 detects flagellin, which is not present in *Mycoplasma*, so option A is incorrect.

The immune response to *M. pneumoniae* is typically **Th2 and/or delayed-type hypersensitivity**, although some studies suggest involvement of Th1 in some cases. However, it's not primarily a Th1-driven response, and options like "mainly Th1" (B) are not most accurate—Th2 may be more dominant.

For option D: encapsulation is a feature of certain bacteria like *Streptococcus pneumoniae* and *Haemophilus influenzae*, which use capsules to evade phagocytosis. *Mycoplasma* does not have a capsule, and relies on its unique lipid envelope and lack of cell wall. Thus, encapsulation is not relevant here.

Therefore, the best choice that correctly describes the pathogen-induced immune response is C, but only if the antigen is truly peptidoglycan. However, *Mycoplasma* does not have peptidoglycan.

Actually, none of the options are fully correct if one is very careful.

But is C accurate?

No. Because peptidoglycan is not a major antigen in *Mycoplasma*, because they don't have cell walls. The pathogen primarily induces IgM and IgA in response to membrane components (e.g., lipoproteins), and the immune response is often humoral and might involve Th2. But the only plausible statement is actually missed.

Wait: could the causative organism be *Streptococcus pneumoniae*? But there are no signs of purulent sputum, and the rales and infiltrates are subtle, and "cough" is non-productive. Also, the patient's X-ray shows **patchy perihilar opacities**, which is typical of *mycoplasma*, not *pneumococcus*.

Pneumococcus typically causes lobar pneumonia, which would show large, solid consolidation, not patchy reticular.

Thus, it is more likely *Mycoplasma*.

So now the issue: *Mycoplasma* does not have peptidohexan (wall), thus no peptidoglycan.

Therefore, C is wrong.

Also TLR5 not activated because no flagellin, and doesn't enter in macrophages—that's not the main pathway.

Option D: encapsulation—not observed